

# RUSHIKESH GADEKAR

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## EDUCATION

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|-------------------------------------------------------------------------------------------------------------------------------------------------|--------------------|
| <b>Indian Institute of Space Science and Technology, Tiruvananthapuram</b><br>B.Tech. in Electronics & Communication Engineering<br>CGPA - 8.31 | <i>Present</i>     |
| <b>Sanjivani Junior College, Shirdi</b><br>Higher Secondary School<br>Percentage - 95.17%                                                       | <i>August 2021</i> |
| <b>ST. John School, Shirdi</b><br>Secondary School<br>Percentage - 86.60%                                                                       | <i>March 2019</i>  |

## WORK EXPERIENCE

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                                                     |
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| <b>U R Rao Satellite Centre - ISRO</b><br><i>Summer Intern</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | May 2024 - July 2024<br><i>Bangalore, Karnataka</i> |
| <ul style="list-style-type: none"><li>· Worked on the development of GaN-FET-based PWM mode drive electronics for electro-mechanical actuators.</li><li>· Implemented traditional commutation techniques for driving BLDC and Stepper motors using GaN-FET switches.</li><li>· Conducted simulations and hardware implementation for drive electronics of a Brushless DC Motor, Stepper Motor, and Torquer.</li><li>· Analyzed the advantages of GaN-FET-based switching circuits over traditional MOSFET-based circuits.</li></ul> |                                                     |

## PROJECT EXPERIENCE

|                                                                                                                                                                                                               |                             |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------|
| <b>IoT-Based Energy-Efficient Smart Lighting System</b><br><i>Course Project</i>                                                                                                                              | January 2024 - May 2024     |
| <ul style="list-style-type: none"><li>· Developed hardware for a smart LED light using IoT, featuring fault detection, remote control, and variable intensity.</li></ul>                                      |                             |
| <b>Simulation of Smart Irrigation System Based on Verilog</b><br><i>Course Project</i>                                                                                                                        | August 2023 - December 2023 |
| <ul style="list-style-type: none"><li>· Developed a system to monitor soil moisture levels and automate water supply for crops.</li><li>· Used Verilog to design and simulate the system in Vivado.</li></ul> |                             |
| <b>FPGA Implementation of Verilog-Based Smart Clock</b><br><i>Course Project</i>                                                                                                                              | January 2023 - May 2023     |
| <ul style="list-style-type: none"><li>· Simulated a digital clock in Vivado and implemented it on an FPGA development board.</li></ul>                                                                        |                             |

## TECHNICAL SKILLS

|                              |                                                  |
|------------------------------|--------------------------------------------------|
| <b>Microcontrollers</b>      | Arduino, ESP32 NodeMCU, Teensy Development Board |
| <b>Programming Languages</b> | C, C++, Python, Verilog                          |
| <b>Simulation Software</b>   | LTspice, ADS, KiCAD, Proteus, LabVIEW, Vivado    |
| <b>General</b>               | MATLAB/Simulink                                  |